

Heaven's Light is Our Guide

Rajshahi University of Engineering & Technology



Department of Mechatronics Engineering **Lab Report**

Course Title : Power Electronics and Drives Sessional
Course Code : MTE 3201
Experiment No. : 04
Experiment Name : Study of Thyristor (SCR) Characteristics.

Date of Experiment : 06-09-2026
Date of Submission : 20-09-2026

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Experiment No: 04

Experiment Name: Study of Thyristor (SCR) Characteristics.

Objectives:

1. To study the gate-triggering characteristics of a thyristor (SCR).
2. To determine the gate trigger voltage (VGT) required to turn on the thyristor.
3. To observe the anode-to-cathode voltage (VAK) before and after triggering.

Introduction:

A thyristor (Silicon Controlled Rectifier, SCR) is a three-terminal, four-layer (PNPN) semiconductor device that behaves as a controlled switch. When the anode is positive with respect to the cathode, the device remains in the forward-blocking (off) state and no significant current flows, even though it is forward biased, until a suitable trigger signal is applied to the gate terminal. As the gate-to-cathode voltage is increased, a corresponding gate current flows; once this reaches the gate trigger voltage/current (VGT/IGT), the thyristor switches into the forward-conduction (on) state. In this state, the anode-to-cathode voltage (VAK) collapses from nearly the full supply voltage to a small forward voltage drop, and the device continues to conduct even if the gate signal is removed, until the anode current falls below the holding current. This experiment studies this triggering behaviour by varying the gate voltage with a potentiometer and observing the corresponding change in VAK on a storage oscilloscope, in order to identify the gate voltage at which triggering occurs.

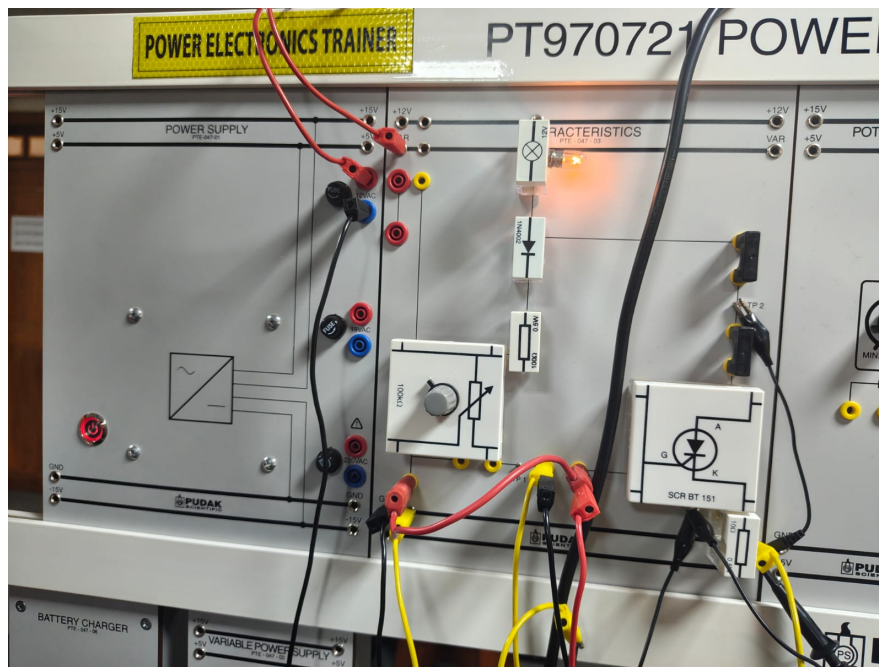


Figure-01: PT970721 POWER ELECTRONIC TRAINER

Required Apparatus:

1. Power supply, PTE-047-01
2. Thyristor characteristics module, PTE-047-03
3. AC lamp, 12 V/2 W
4. Diode, 1N4002
5. Potentiometer, 10 k Ω
6. Thyristor (SCR), BT151
7. Digital multimeter
8. Storage oscilloscope
9. Connecting leads

Principle:

The circuit consists of a 12 V AC source feeding a lamp (12 V/2 W, used as a current-limiting indicator) in series with a diode (1N4002) and a resistor R1, which together provide a rectified reference voltage. A potentiometer R2 is used to derive an adjustable gate voltage, which is applied to the gate of the thyristor (BT151) through the small resistor R3. The gate-to-cathode voltage developed across R3 (UGT) and the anode-to-cathode voltage of the thyristor (UAK) are observed simultaneously on the two channels of the oscilloscope (CH1 = UAK, CH2 = UGT).

As the potentiometer R2 is adjusted, the gate voltage UGT increases from zero. While UGT remains below the gate trigger voltage of the thyristor, the device stays in the forward-blocking state and UAK remains high (close to the peak of the supply voltage appearing across the device), i.e. the thyristor is "not triggered". Once UGT reaches the minimum gate trigger voltage required for the device, the thyristor turns on, its anode-to-cathode voltage collapses to a small forward conduction drop, and the device is said to be "triggered". By recording UGT and the corresponding UAK for several settings of the potentiometer, the gate-triggering characteristic of the thyristor can be studied and the approximate gate trigger voltage of the device can be identified.

Procedure:

- 1) The required power, thyristor-characteristics, and load panels were connected according to the circuit diagram provided in the manual.
- 2) The AC supply, lamp, diode D1, resistor R1, potentiometer R2, resistor R3, and the thyristor BT151 were connected in series/parallel as shown in the circuit diagram.
- 3) The oscilloscope was connected with CH1 across the anode-cathode terminals (to observe UAK) and CH2 across R3 (to observe the gate voltage UGT).
- 4) The potentiometer R2 was set to its minimum position, and the circuit was switched on.
- 5) The potentiometer was varied gradually, and the gate voltage UGT together with the corresponding anode-cathode voltage UAK was recorded for each setting.
- 6) The point at which the thyristor changed from the "not triggered" (blocking) state to the "triggered" (conducting) state was noted from the sudden change in UAK.
- 7) The observed values of UGT and UAK were recorded in the data table for different potentiometer settings.

Data Table:

No.	VGT (Gate Trigger Voltage)	VAK (Anode-Cathode Voltage)	Remarks
1	26 mV	3.64 V	Not triggered
2	182 mV	2.90 V	Triggered
3	224 mV	3.68 V	Triggered
4	186 mV	2.06 V	Triggered
5	184 mV	3.20 V	Triggered

Note: CH1 was connected to measure UAK and CH2 was connected to measure UGT, as shown in the circuit diagram. Only the first setting (No. 1) failed to trigger the thyristor; for all subsequent, higher settings of the gate voltage the thyristor was found to be successfully triggered.

Result:

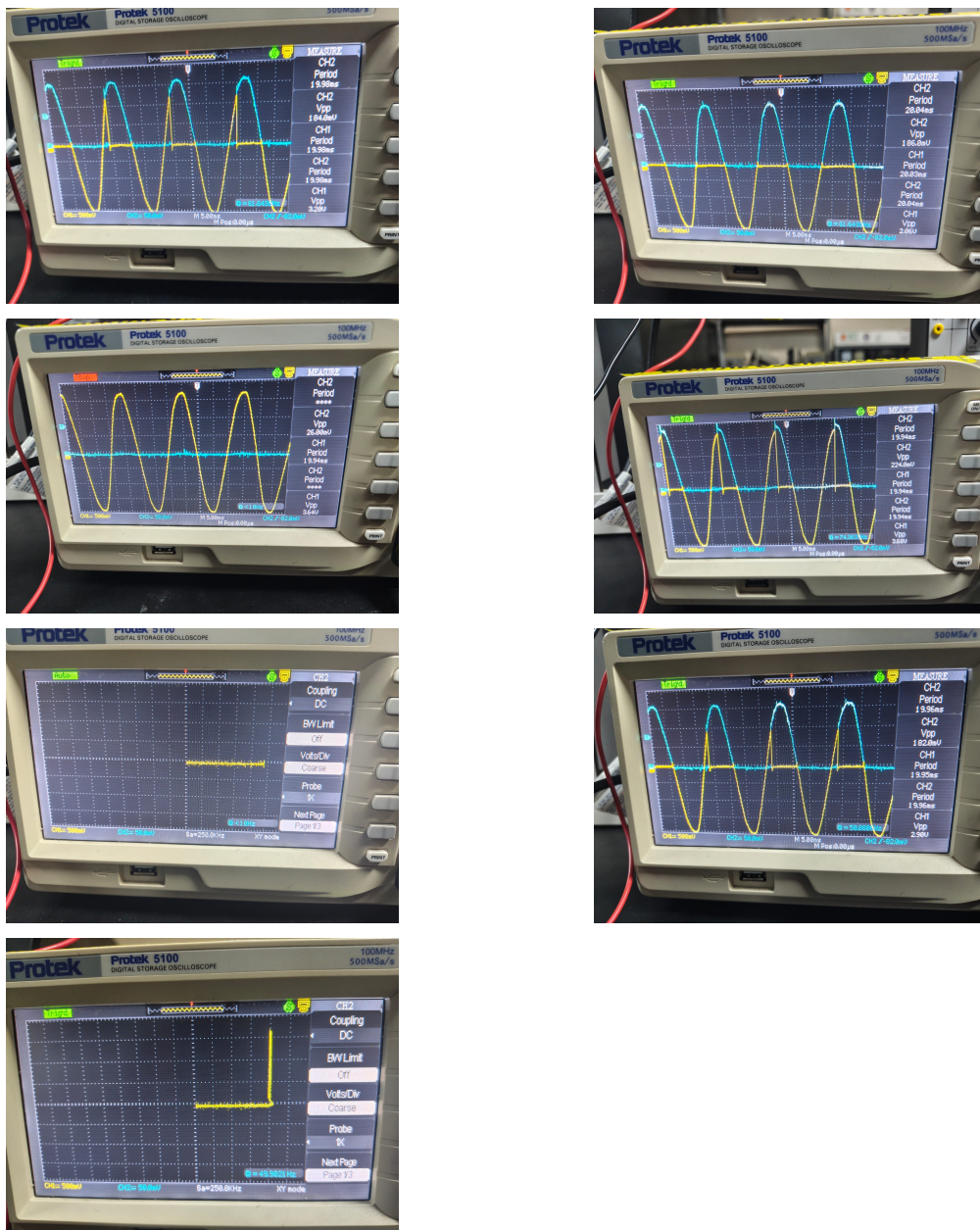


Figure-02: Oscilloscope Graph in Various Phase of the Experiment



Figure-03: DC Variable Power Supply Unit Used in the Experiment

Discussion:

The gate-triggering characteristic of the thyristor (SCR BT151) was studied by varying the gate voltage U_{GT} with a potentiometer and observing the resulting anode-to-cathode voltage U_{AK} on a storage oscilloscope. At the lowest gate-voltage setting ($U_{GT} = 26 \text{ mV}$), the gate voltage was too small to trigger the device, so the thyristor remained in its forward-blocking state and U_{AK} stayed high, confirming that the device was "not triggered". As the potentiometer was increased further, the gate voltage rose into the range of about 180-224 mV, at which point the thyristor was consistently triggered into conduction for every setting tested, and U_{AK} was observed to fall to a lower value each time, consistent with the device switching into its on-state. The small variations in U_{AK} among the triggered readings (2.06 V-3.68 V) are attributed to the exact instant within the AC half-cycle at which triggering occurred, as well as to the loading effect of the lamp and diode branch and to normal measurement and reading tolerances. Overall, the experiment confirmed that the thyristor requires a minimum gate voltage of roughly 150-200 mV under this circuit configuration before it latches into conduction, and that once triggered, the device continues to conduct with a comparatively small anode-to-cathode voltage drop.

Conclusion:

The gate-triggering characteristics of a thyristor (SCR) were successfully studied. It was observed that the thyristor remains in the forward-blocking state and does not conduct as long as the gate voltage is below a certain minimum (gate trigger) value; once this value is exceeded, the device switches rapidly into the forward-conduction state, and the anode-to-cathode voltage falls sharply. The experiment verified the basic switching behaviour of the thyristor and provided a practical estimate of its gate trigger voltage.

References:

[1] Lab manual

[2] M. H. Rashid, *Power Electronics: Circuits, Devices, and Applications*, 4th ed. Pearson, 2014.

[3] N. Mohan, T. M. Undeland, and W. P. Robbins, *Power Electronics: Converters, Applications, and Design*, 3rd ed. John Wiley & Sons, 2003.

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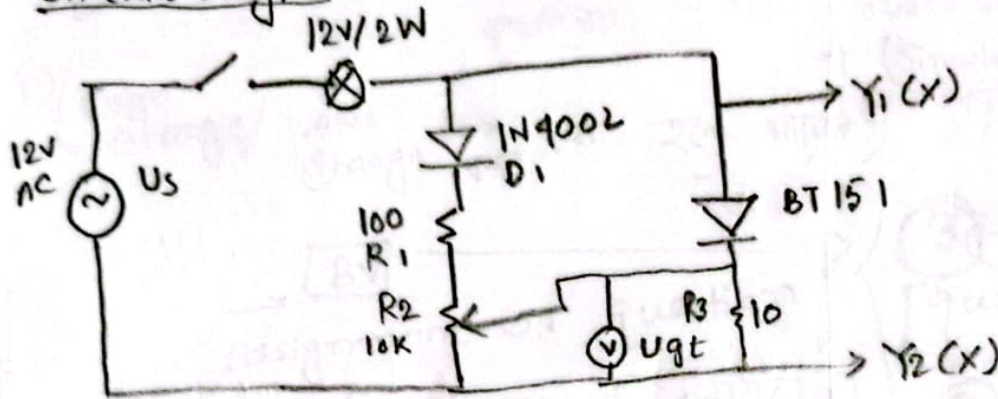
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Revised
 06.09.20

Circuit diagram



Data table

No	$U_{GT} (V)$	$U_{AK} (V)$
1	26mV	3.64V
2	182mV	2.98V
3	224mV	3.68V
4	186mV	2.06V
5	184mV	3.20V

CH1 : U_{AK}
 CH2 = U_{GT}

Not triggered.